



Pressure and flow patterns during multiple-order vertical vortex-induced vibrations of a 6:1 rectangular cylinder

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ABSTRACT

Rectangular cross-sections have been typically used to investigate vortex-induced vibration (VIV) mechanism of bluff bodies, which provides important reference for wind-resistant design of engineering structures, such as high-rise buildings and long-span bridges. A numerical simulation was conducted in order to examine the multiple-order vertical VIV mechanism of a 6:1 rectangular cylinder, and the surface wind pressure distribution characteristics and surrounding flow field were particularly focused. This rectangular cylinder has two VIV lock-in regions. According to the surface pressure distribution characteristics, it is deduced from the proposed simplified vortex theory that there are significant vortices drifting on the upper and lower surfaces of the rectangular cylinder, and the vortex experiences approximate two and one vibration period drifting from the leading edge to the trailing edge in the first and second lock-in region, respectively. More characteristics of the flow field are revealed by dynamic mode decomposition (DMD) method. The front three non-zero DMD mode frequencies are about 1–3 times of the natural frequency of the rectangular cylinder, respectively. Multiple vortices are also observed on the upper and lower surfaces in the high-order DMD modes, especially in the second lock-in region, indicating the importance of vortex drifting phenomenon in the occurrence of VIV. This investigation provides a deeper viewpoint combining surface pressure distribution characteristics and surrounding flow patterns, which is of important reference for revealing the VIV mechanism of wind-sensitive structures with bluff body cross-sections.

1. Introduction

Vortex-induced vibration (VIV), being to typical wind induced vibration, often occurs on long span bridges at low wind velocity range (Ge et al., 2022; Zhao et al., 2022). It appears to in vertical or torsional directions, and affects the safety and comfort of drivers and pedestrians, or reduces the fatigue life of structures. It is of great importance to reveal the VIV mechanism of structures with elongated bluff body configurations and suppress its occurrence. Rectangular cylinders are commonly used to investigate the VIV mechanism of bluff bodies, which provide important reference for wind-resistant design of engineering structures, such as long-span

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bridges and high-rise buildings (Bruno et al., 2014; Zhang et al., 2020).

It is well-known that the vortex shedding from short bluff cylinders (such as circular or square-section cylinders) is characterized by Karman vortex pattern, however, the vortex shedding from elongated bluff cylinders, such as rectangular cylinders with large aspect ratios (the ratio of width to height), can be characterized by impinging shear-layer instability, where a single separated shear layer becomes unstable in the presence of a sharp downstream corner (Nakamura and Nakashima, 1986; Nakamura et al. 1996). Nakamura et al. (1991) studied the vortex shedding phenomenon around rectangular cylinders with aspect ratios of 3–16 by wind tunnel tests, which found that the Strouhal number would approximately increase stepwise in an integral multiple of 0.6 with the increase of the aspect ratio. A further investigation by Ohya et al. (1992) illustrated this phenomenon of Strouhal number is closely related to the number of vortices formed on each surface. Kubo et al. (1992) conducted wind tunnel tests and flow visualization for a rectangular cylinder, and revealed that the occurrence of VIV as well as its motion (torsional or vertical) depends on the number of vortices drifting on the surfaces. Mills et al. (2003) elucidated that the leading-edge vortices always pass the trailing edge at the same phase regardless the aspect ratio. Nguyen (2018) explained the VIV mechanism of a 5:1 rectangular cylinder, and revealed that the VIV is triggered by the motion-induced leading-edge vortex, during which a strongly correlated flow feature close to the trailing edge is responsible for an increase in the structural response. Zhang et al. (2023) conducted a large-eddy simulation of the unsteady flows around rectangular prisms with aspect ratios of 3–12, which found that large-scale leading-edge vortices are formed by the coalescence of Kelvin–Helmholtz rollers, and the phase difference between the leading edge and trailing edge vortices is the critical factor promoting a stepwise increase in the Strouhal number. These investigations demonstrate that the VIV of elongated rectangular cylinders is closely related to the separated vortices generated at the leading edge and their drifting characteristics on the surface, rather than just caused by the Karman-type vortices alternately shedding from the trailing edge. The number and drifting characteristics of the vortices on the surfaces, as well as the mutual relationship between the leading edge vortices and trailing edge vortices play an important role in the occurrence of VIVs. However, the flow pattern characteristics as well as VIV mechanism of rectangular cylinders still has not been entirely clarified by now.

Multiple-order VIVs has been observed in elongated bluff bodies (Komatsu and Kobayashi, 1980; Shiraishi and Matsumoto, 1983; Nakamura and Nakashima, 1986; Nguyen et al., 2018) and bridge decks in previous researches (Hu et al., 2022; Wu et al., 2017). Shiraishi and Matsumoto (1983) claimed that the separated vortices from the leading edge ought to reach to the trailing edge after n vibration periods for vertical VIV or after $(2n-1)/2$ vibration periods for torsional VIV, in which n can be taken as different positive integers for multiple-order VIVs. Nguyen et al. (2018) found that in the two lock-in regions of a 5:1 rectangular cylinder, there are two and one vortices on the surfaces, respectively. Hu et al. (2022) found that the three lock-in regions of a streamlined closed-box girder are caused by the leading-edge vortices with different drifting periods and wake vortices, respectively. Wu and Kareem (2017) attributed the multi-lock-in regions of a wide flat box girder to the phenomenon that three nascent vortices originating at the pedestrian guardrails and inspection rails shed to near-wake through different pathways with different frequencies. However, this explanation is not suitable for the multiple-order VIVs of rectangular cylinders without additional facilities. The differences between multiple-order VIVs of rectangular cylinders have not been fully elucidated so far, especially on the aspect of surface pressure distribution characteristics and surrounding flow pattern. Further investigations and validations are needed.

The flow field characteristics influence the wind pressure on the structure surface and can be reflected from its distribution trend (Komatsu et al., 1980; Hu et al., 2022, 2024), and surface pressure measurement is one of the most important and effective means to reveal the VIV mechanism of structures (Hu et al., 2018; Laima et al., 2018; Liu et al., 2021). Komatsu et al. (1980) found that separated vortices from the leading edge could produce regular fluctuating pressure on a rectangular cylinder surface, and lead to VIVs. Shang et al. (2019) and Zhou et al., 2024 found that the chamfered-corner ratios and spacing ratios play important roles in the qualitative change of the flow structures of two tandem cylinders, at the same time, the aerodynamic force also change. Hu et al. (2021, 2022 and 2024) studied the mapping relationship between surface wind pressure distribution and vortex drifting pattern, and established a simplified vortex model about the VIV mechanism of a streamlined closed-box girder. This method of building simplified vortex model needs to be further validated and developed on structures with different shapes. For rectangular cylinders, investigations about the surface pressure distribution characteristics and the mapping relationships between the surface pressure and the flow field are not detail enough. These investigations would be helpful to reveal the VIV mechanism in depth, as well as further validate and develop the simplified vortex model method. Surface wind pressure is usually obtained by wind tunnel tests. However, the flow pattern around structure cannot be directly obtained without expensive flow visualization equipment such as particle image velocimetry (PIV) system in wind tunnel tests, which makes it difficult to reveal VIV mechanism from flow field characteristics. Computational fluid dynamics (CFD) provides an alternative way to investigate VIVs, which uses numerical simulation methods to obtain the flow field characteristics and corresponding surface pressure as well as the structure displacement at the same time. As a result, it is possible to further reveal the VIV mechanism from the viewpoint which combines surface pressure distribution characteristics and surrounding flow patterns, as well as further verify and establish the mapping relationship between surface pressure and key flow field characteristics.

Dynamic mode decomposition (DMD) method proposed by Schmid (2010) is a useful way to decompose the original flow into a series of modes with corresponding frequency, so that spatiotemporal coherent structures in periodic flows can be identified. This method has become an effective way to investigate the flow mechanism for cavity and cylinders (Lusseyran et al., 2011; Guéniat et al., 2014; Li et al., 2019; Zhang et al., 2023). Zhang et al. (2014) conducted a comprehensive comparison of proper orthogonal decomposition (POD) and DMD algorithms. The result showed that the DMD method could clearly separate each flow structure in the spatial and spectral senses, while the POD mode corresponding to the desired flow structure is contaminated by the other uncorrelated flow structures. Luo and Kareem (2021) also found that POD tends to mix the temporal frequencies in the identified modes, while DMD provides single-frequency eigenmodes. Besides, the DMD modes can capture the evolution of aerodynamic features, such as convection

of vortices (or vortex tubes) and other structures. For rectangular cylinders, Zhang et al. (2023) conducted DMD to investigate the flow field characteristics and revealed the physical mechanisms governing the global instability of the flow. However, further investigations need to be done to reveal the VIV mechanisms and show the differences between multi-order VIVs.

In the present study, a numerical simulation is conducted for a 6:1 rectangular cylinder, to investigate its multiple-order vertical VIV mechanism from the viewpoint combining surface pressure distribution and surrounding flow field characteristics. Besides, DMD method is used to provide further information about the flow patterns under VIVs. The rest of this paper is organized as follows. The details about numerical simulation setup and results are introduced in Section 2. In Section 3, the wind pressure and aerodynamic force distribution characteristics on surfaces are presented. According to these, the flow patterns around the rectangular cylinder are inferred combining the simplified vortex model (Hu et al., 2021 and 2022). And in Section 4, the flow field from the numerical simulation is shown to verify the correctness of this inference, and DMD method is used to illustrate the flow pattern corresponding to each key frequency and the VIV mechanism is further explained. Finally, the main conclusions are summarized in Section 5. This investigation not only reveals the differences between the mechanism of the two vertical VIV lock-in regions of the 6:1 rectangular cylinder, but also provides an alternative viewpoint which combines the surface pressure characteristics and the surrounding flow field to reveal VIV mechanism. This method has important reference value for the mechanism investigations of more complex structures with typical bluff body configurations.

2. Numerical simulation

2.1. Computation setup

The theoretical basis of CFD is the Navier-Stokes equations, including conservation equations for mass, momentum and energy. The governing equations are solved by ANSYS FLUENT, a computational software based on finite volume method. Two-dimensional (2D) Reynolds-averaged Navier-Stokes equations (RANS) and SST $k-\omega$ turbulence model are used, which are acceptable in terms of computation speed and simulation accuracy.

The rectangular section used for calculation is shown in Fig. 1. with a height of $D = 67$ mm and a width of $B = 400$ mm, the aspect ratio is about 6. The vertical natural frequency of the rectangular cylinder is 6.71 Hz, the vertical damping ratio is 0.54 %, and the mass per unit is 6.17 kg/m. The streamwise and transverse dimensions of the computational domain are set as 30B and 20B respectively, so that the block ratio is 0.8 %. The distance from the middle of the rectangular cylinder to the velocity inlet boundary and pressure outlet boundary are about 10B and 20B, respectively, as shown in Fig. 2. The top and bottom boundaries are set as symmetric boundary, and the rectangular surface is set as no-slip wall boundary.

The calculation domain is divided into three parts: the rigid body area, the dynamic-mesh area and the static-mesh area (Fig. 2). A boundary layer exists around the wall of the rectangular cylinder (Fig. 3), which contains 30 layers of structured grids. The height of the first layer grids near the wall is $<0.0005D$, and the growth rate is 1.12. Triangular unstructured grids are arranged around the boundary layer as transitions. These boundary layer and transition grids are set as the rigid body area. The dynamic-mesh area is outside the rigid body area. The grids in this area are all triangular unstructured grids and update with the movement of the rectangular cylinder. The static-mesh area is the outermost region, and the grids in this region are all structured grids. In general, the grids gradually become denser as the position approaches the wall of the rectangular cylinder. The total number of the grids is 49,936, and the maximum dimensionless wall distance y^+ is 0.64.

User-design function (UDF) is used to achieve the simulation of vertical VIVs. First, the lift force acting on the rectangular cylinder is extracted, and Newmark- β method is used to solve the vertical displacement response. The motion of the rectangular cylinder is transferred to the FLUENT solver to update the position of dynamic meshes. The pressure-velocity coupling is achieved by the SIMPLEC algorithm. The spatial discretization is set as Least Squares Cell Based for gradient, PRESTO for pressure, QUICK for momentum and turbulent kinetic energy, and Second Order Upwind for turbulent dissipation rate. The First Order Implicit scheme is selected for the temporal discretization. When calculating, the iteration time step is set to 0.001 s, and the convergence residual is defined as 1×10^{-6} . The maximum Reynolds number in the simulation is about 1.9×10^5 .

2.2. Mesh sensitivity verification

To evaluate the influence of mesh density and number on the simulated flow field, a mesh sensitivity verification was conducted by performing the static simulation. This study involves three different mesh schemes, and the main differences between them are the density and number of boundary layer grids and the triangular grids around them. The basic information of the three mesh schemes

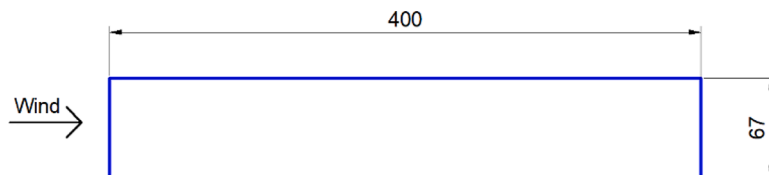


Fig. 1. Layout of a 6:1 regular section (unit: mm).

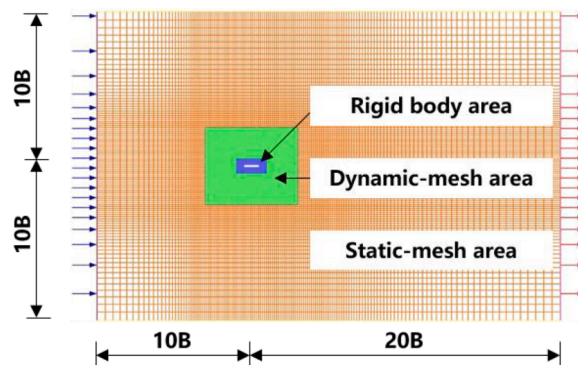


Fig. 2. Computational domain and mesh refinement.

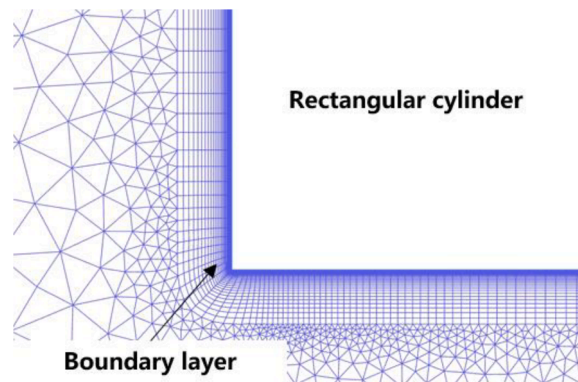


Fig. 3. Boundary layer around rectangular cylinder.

and the calculation results of mean drag coefficient (C_{d-avg}) and Strouhal number are shown in Table 1.

According to Table 1, there is no significant differences between the results of the three mesh schemes. Considering both the accuracy and calculation time, Mesh 1 is selected as the scheme using in thereafter simulation of VIVs. In former investigation of a similar 6:1 rectangular cylinder conducted by Shang et al. (2015), the S_t number is about 0.60, which are very close to the results in Table 1. In other words, the results of simulation are reliable with the satisfactory precision.

2.3. Calculation results

A uniform flow velocity is set at the velocity inlet boundary. The displacement responses of the rectangular cylinder under different wind velocities are calculated, and the results are shown in Fig. 4. In the figure, the ordinate is the normalized amplitude, which is the ratio of the vertical displacement to the height of the rectangular cylinder. The abscissa is the reduced velocity, $V_r = U/f_h D$, where U denotes the oncoming wind velocity, f_h denotes the vertical natural frequency, and D denotes the height of the rectangular cylinder. Two lock-in regions exist for the 6:1 rectangular cylinder, whose reduced velocity ranges are approximately 5.12–8.01 and 8.90–15.13, respectively. The maximum amplitude of the second lock-in region is about 0.068 (the corresponding reduced velocity is $V_{r2}=13.75$), which is >2 times of the maximum amplitude of the first lock-in region 0.030 (the corresponding reduced velocity is $V_{r1}=6.23$). The corresponding Reynolds number of the two peak amplitude points are 7.6×10^4 and 1.7×10^5 , respectively. Similar phenomenon of multiple-order VIVs on rectangular cylinders were found by Komastu and Kobaysshi (1980), Nakamura and Nakashima (1986) and Nguyen et al. (2018).

Shang et al. (2015) conducted wind tunnel test and CFD simulation investigation on a 6:1 rectangular cylinder, which owned the same width and height with present simulation (Fig. 1), and the length of the wind tunnel test model was about 1540 mm. The vertical

Table 1

Computational grids in mesh sensitivity study.

Mesh	Number	Max y^+	C_{d-avg}	Strouhal number S_t
Mesh 1	49,936	0.64	0.948	0.59
Mesh 2	63,704	0.62	0.954	0.60
Mesh 3	75,432	0.43	0.945	0.60

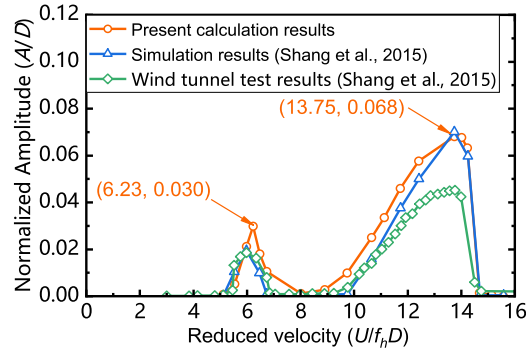


Fig. 4. Multiple-order VIV response and validation.

natural frequency, the damping ratio, and the mass per unit of the wind tunnel test model and simulation set of Shang et al. (2015) were all identical with the simulation set in present study. The results of Shang et al. (2015) are also shown in Fig. 4. In the aspect of lock-in wind velocity, the CFD simulation results of present study correspond well with the CFD simulation and wind tunnel test results of Shang et al. (2015). However, for VIV amplitude, the two CFD simulation results agree well, but larger than the wind tunnel test results. The spring suspended system used in wind tunnel tests often shows nonlinearity on damping ratio (Gao et al., 2015), and the damping ratio varies with the VIV amplitude in the wind tunnel test, which may lead to errors in the measurement of damping ratio and cause the differences on VIV amplitude between CFD simulation and wind tunnel test results. Besides, the 3D flow pattern (Bruno et al., 2010) and spanwise correlation (Matsumoto et al., 2003) may also influence the VIV amplitude. The main purpose of this investigation is to reveal the differences on wind pressure distribution trend and flow pattern between the two lock-in regions, the exact values of VIV amplitude are hardly included. Due to the correspondence in VIV lock-in velocities, it is considered acceptable to use the current numerical simulation results to conduct the following investigations on the surface wind pressure distribution trend and vortex drifting pattern.

3. Surface pressure characteristics

The surface pressure of the 6:1 rectangular cylinder is extracted to investigate the VIV mechanism of the two lock-in regions and the differences between them. The wind pressure coefficient on each point on the rectangular cylinder surface is defined as follows:

$$C_{pi}(t) = \frac{p_i(t)}{1/2\rho U^2} \quad (1)$$

where $C_{pi}(t)$ is the non-dimensional wind pressure coefficient; $p_i(t)$ is the time series of the wind pressure measured at the i^{th} point; ρ is the air density (1.225 kg/m^3). Positive wind pressure coefficient means that the wind pressure acting towards the structure surface, while negative wind pressure coefficient means that the wind pressure acting away from the structure surface.

The mean pressure coefficient is the time average of the wind pressure coefficient, $C_{pi}(t)$:

$$C_{pi,\text{mean}} = \frac{1}{N} \sum_{n=1}^N C_{pi}(t_n) \quad (2)$$

The fluctuating pressure coefficient is defined by Eq. (3), which is a reflection of wind pressure pulsation amplitude and the magnitude of the dynamic wind load:

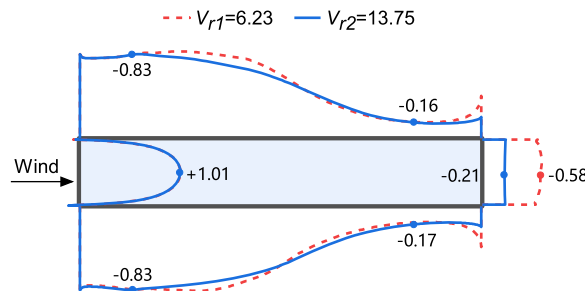


Fig. 5. Mean pressure coefficient at peak amplitude points of two lock-in regions during VIVs.

$$C_{pi, std} = \sqrt{\frac{1}{N} \sum_{n=1}^N (C_{pi}(t_n) - C_{pi, mean})^2} \quad (3)$$

The distribution of the mean pressure coefficient and fluctuating pressure coefficient at the two peak amplitude points ($V_{r1}=6.23$ and $V_{r2}=13.75$) on the rectangular cylinder surface are shown in Fig. 5 and Fig. 6, respectively. In Fig. 5, positive mean pressure coefficient is marked by lines inside the rectangular cylinder surface, while negative value is marked by lines outside the surface. According to Eq. (3), the fluctuating pressure coefficient can only be positive, and it is marked by lines outside the rectangular surface in Fig. 6.

Both the two coefficients are symmetrically distributed along the upper and lower surface. The mean pressure coefficient is only positive on the windward side of the rectangular cylinder, and turns to negative near the leading edge corners. This phenomenon means that the airflow separates after passing the leading edge corners, and then leading edge separated vortices generate. On the leading edge and the upper and lower surfaces, the mean pressure coefficients are similar at the two peak amplitude points, but on the trailing edge, the absolute values of mean pressure coefficients under V_{r1} is significantly larger than those under V_{r2} . The distribution trend of mean pressure coefficient corresponds well with the investigation of Li et al. (2021).

In terms of fluctuating pressure coefficient, the distribution characteristics at the two peak amplitude points are obviously different, implying different VIV excitations and sustained mechanisms. Under reduced velocity V_{r1} , there are two peak values and two valley values on the upper or lower surfaces, respectively, which indicates that there are two vortices on each surface (Hu et al., 2022; Komastu et al., 1980). On the trailing edge, the fluctuating pressure is relatively larger near the corners and smaller in the middle part. However, under reduced velocity V_{r2} , the fluctuating pressure only reaches its maximum near the location of about $3/4B$, indicating that there may be only one vortex on each surface. The value on the trailing edge is uniformly distributed and quite small, which is also different with the first lock-in region, indicating the differences between the flow patterns around the trailing edge of the two lock-in regions.

Besides the fluctuating pressure, the correlation between the distributed aerodynamic force and general VEF should also be considered when describing the wind pressure distribution characteristics (Hu et al., 2018). The correlation between the distributed aerodynamic force and the general VEF can be expressed by the correlation coefficient defined as follows:

$$\rho_i = \frac{\text{cov}(F_{aero}(t), F_i(t))}{\sqrt{D(F_{aero}(t))} \sqrt{D(F_i(t))}} \quad (4)$$

where $F_{aero}(t)$ is the non-dimensional general VEF, that is, the non-dimensional overall lift coefficient of the rectangular cylinder calculated synchronously with the surface pressure; $F_i(t)$ is non-dimensional distributed aerodynamic force on different position of the surface, which is the projection of the wind pressure coefficient $C_{pi}(t)$ in the vertical upward direction. $D(F_{aero}(t))$ and $D(F_i(t))$ denote the variances of $F_{aero}(t)$ and $F_i(t)$, respectively, and $\text{cov}(F_{aero}(t), F_i(t))$ denotes the covariance between $F_{aero}(t)$ and $F_i(t)$.

For vertical VIVs, the influence of the distributed aerodynamic force on the general VEF can be described quantitatively by the product of the correlation coefficient and the fluctuating pressure coefficient, which is defined as contribution value (Hu et al., 2018):

$$C_{aero-i} = C_{pi, std} \times \rho_i \quad (5)$$

The distribution characteristics of the correlation coefficient and contribution value at the peak amplitude points of the two lock-in regions are shown in Fig. 7 and Fig. 8, respectively. Positive correlation coefficients and contribution values mean that the distributed aerodynamic forces on these points contribute positively to the general VEF, and negative values mean that distributed aerodynamic forces on these points mitigate the general VEF. In these figures, positive values are marked by lines inside the rectangular outlines and negative values are marked by lines outside the rectangular outlines.

According to Fig. 7, the positive and negative values of correlation coefficient alternately appear on the surfaces of the rectangular cylinder, and the distribution characteristics on the upper and lower surfaces are approximately the same. Under V_{r1} , there are two positive value zones and two negative value zones on each surface (neglecting the small part after the leading edge corners), while there is only one positive value zone and one negative value zone on each surface under V_{r2} . The contribution values in Fig. 8 show similar distribution characteristics with the correlation coefficients, that is, positive and negative values alternately appear on the surfaces. At the two peak amplitude points, the distribution trends of contribution values vary with the coordinate in the form of simple

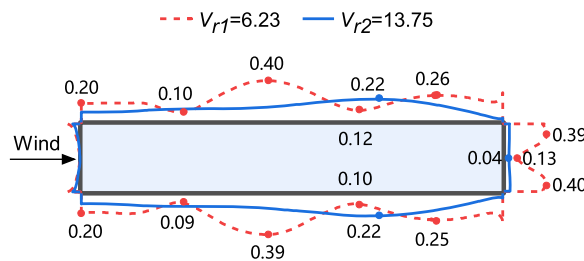


Fig. 6. Fluctuating pressure coefficient at peak amplitude points of two lock-in regions during VIVs.

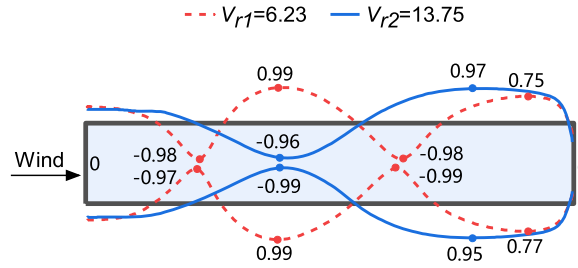


Fig. 7. Correlation coefficient at peak amplitude points of two lock-in regions during VIVs.

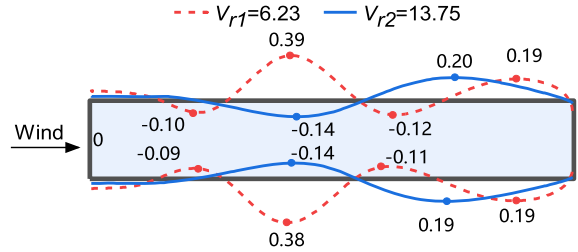


Fig. 8. Contribution value at peak amplitude points of two lock-in regions during VIVs.

harmonic curves of two and one period, respectively, and the corresponding wavelengths of the simple harmonic curves are approximately equal to 1/2 times and once of the width of the rectangular cylinder. According to the investigation of Hu et al. (2022) about the multiple-order VIV mechanisms of a streamlined closed-box girder and the simplified vortex model (Hu et al., 2021), this phenomenon of contribution value distribution indicates that there are two vortices drifting on each surface in the first lock-in region and one vortex in the second lock-in region. These vortex drifting phenomenon can be further confirmed from the distribution characteristics of the phase lags between the distributed aerodynamic force at each point on the surface and the general VEF.

The phase lag between the general VEF and the distributed aerodynamic force at each point on the surface at different frequencies can be obtained by calculating their cross-spectral density, and its distribution at the natural frequency of the rectangular cylinder is shown in Fig. 9. Except the small parts around the leading edge and the trailing edge, the phase lags on the upper and lower surfaces shows an approximately linear increasing trend, with different increasing rate at the two peak amplitude points. This indicates that there are vortices drifting on the upper and lower surfaces, and affecting the aerodynamic forces on each point from upstream to downstream in turn (Hu et al., 2022). From the leading edge to the trailing edge, the phase lag variations under V_{r1} and V_{r2} are about 618° and 340° , which are close to 720° and 360° , respectively. This means that under V_{r1} , it takes about two vibration periods for the

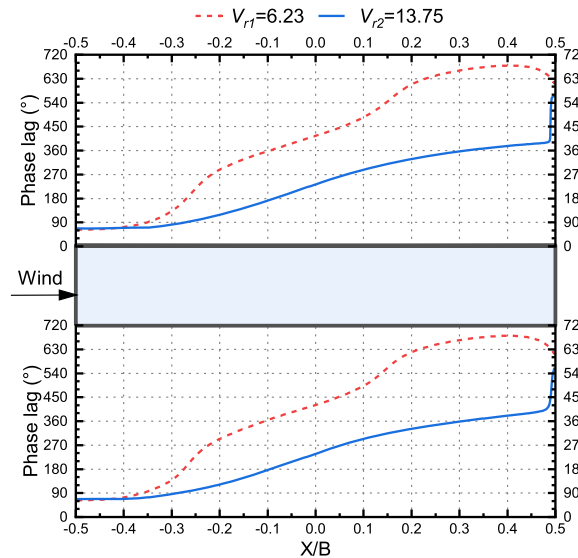


Fig. 9. Phase lag at two peak amplitude points of the two lock-in regions during VIVs.

leading edge vortex to drift to the trailing edge, and under V_{r2} , the corresponding time is about one vibration period. [Nguyen et al. \(2018\)](#) also found similar phenomenon by wind tunnel test and CFD simulation on a 5:1 rectangular cylinder: in the two lock-in regions, there are two and one vortex on the surfaces, respectively. This phenomenon of periodical vortex drifting is closely related to the multiple-order VIVs of rectangular cylinders.

4. Flow field characteristics

4.1. Simplified vortex model and original flow field

According to the distribution of contribution value and phase lag, the flow field characteristics around the rectangular cylinder at the two peak amplitude points are deduced combining the simplified vortex model theory by [Hu et al. \(2021 and 2022\)](#), as shown in [Fig. 10\(a\)](#) and [Fig. 11\(a\)](#). The vortices in these figures are simplified vortices instead of real vortices in actual flow fields. These simplified vortices generate after the leading-edge corners, drifting to the trailing edge with a constant velocity. The vortices on the upper surface generate upward forces, and the vortices on the lower surface generate downward forces, as shown by the solid black

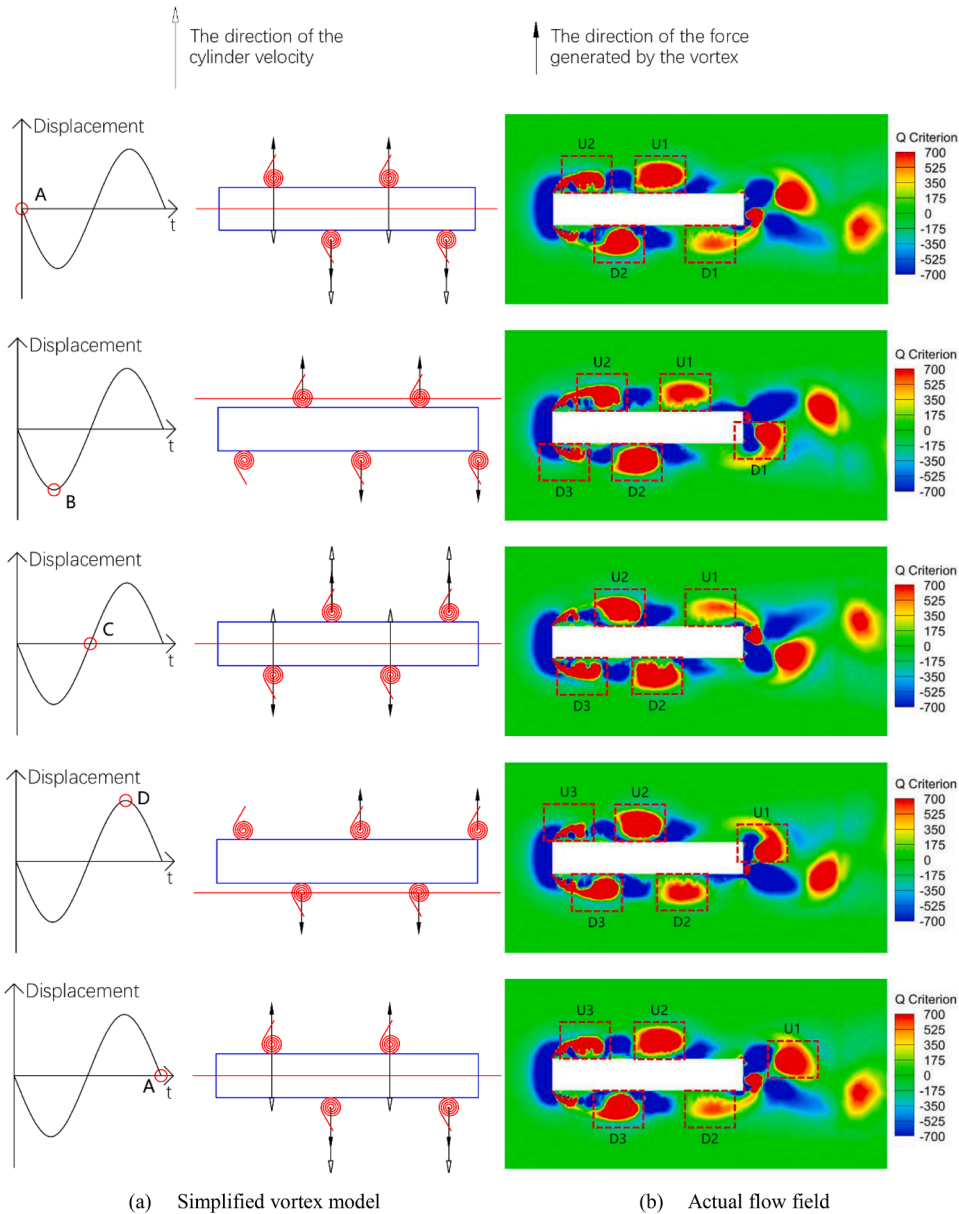


Fig. 10. Flow field at peak amplitude point of the first lock-in region.

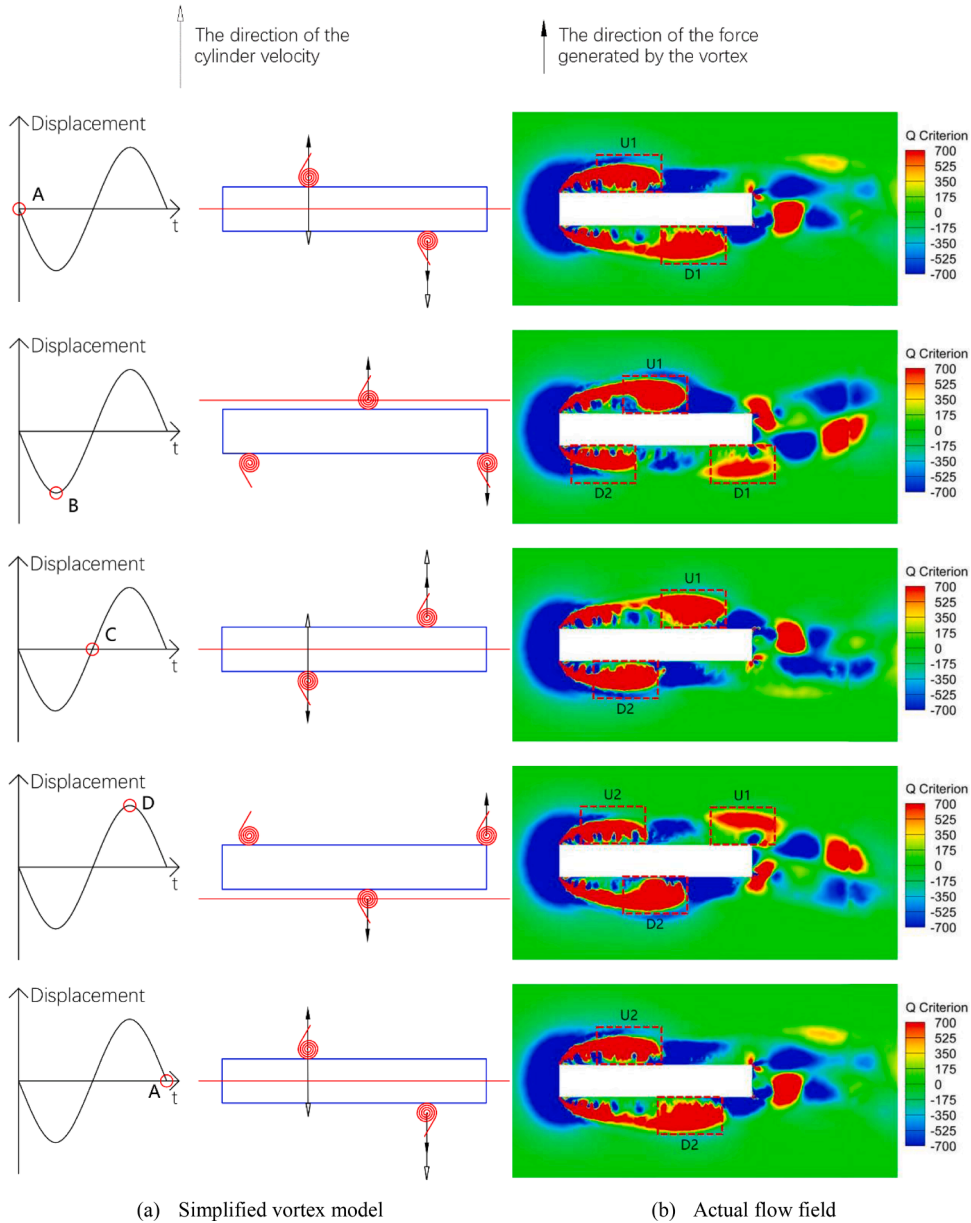


Fig. 11. Flow field at peak amplitude point of the second lock-in region.

arrows in the Fig. 10(a) and Fig. 11(a). The white arrows in the figures indicate the velocity direction of the rectangular cylinder. When the force generated by the vortex is in the same direction with structure movement velocity, the vortex does positive work, otherwise, the vortex does negative work. The actual flow field obtained by CFD simulation is shown in Fig. 10(b) and Fig. 11(b) for comparison and verification of the simplified vortex model.

In the first lock-in region, there are two vortices drifting on the upper and lower surfaces, respectively. In a vibration period, when the rectangular cylinder moves to the equilibrium position and is about to move downward, it owns the maximum downward motion velocity (position A in Fig. 10(a)). At this time, the forces generated by the upper surface vortices (U1 and U2) do negative work on the structure, locating on the positions corresponding to the maximum negative contribution values; the forces generated by the lower surface vortices (D1 and D2) do positive work, locating on the positions corresponding to the maximum positive contribution values. All these vortices drift downstream when the rectangular cylinder vibrates. When the rectangular cylinder moves downward to its maximum displacement (position B), its velocity is 0, and all the vortices do no work. Vortex D1 sheds from the lower surface and interacts with the trailing edge, causing the large fluctuating pressure on this position. At the same time, vortex D3 generates after the leading-edge corner of the lower surface, so that there are still two vortices on the lower surface. The rectangular cylinder moves upward to position C, the equilibrium position, where the structure owns the largest upward velocity. The upper surface vortices do

positive work, drifting to the positions with the maximum positive contribution values; and the lower surface vortices do negative work, drifting to the positions with the maximum negative contribution values. Then the rectangular cylinder moves upward to position D, where the displacement reaches the maximum positive value, and the movement velocity is zero. The upper surface vortex U1 sheds from the trailing edge corner, and interact with the trailing edge, while the upper surface vortex U3 generates after the leading-edge corner. The rectangular cylinder moves downward back to position A, and a new period begins. U2 and D2 move to the original positions of U1 and D1, respectively; and U3 and D3 move to the original positions of U2 and D2, respectively. This means that each vortex travels a distance equal to about half the surface length in one period, in other words, it takes two periods for one vortex to drift from the leading-edge to the trailing edge.

As for the second lock-in region, there is only one vortex drifting on the upper and lower surface, respectively. It takes about one period for the vortices from the leading-edge drifting to the trailing edge. On position A, the upper surface vortex (U1) locates at the position corresponding to the maximum negative contribution value and does negative work; the lower surface vortex (D1) locates at the position corresponding to the maximum positive contribution value and does positive work. When the rectangular cylinder moves downward to location B, the lower surface vortex D2 generates after the leading-edge corner, at the same time, the lower surface vortex D1 sheds from the trailing edge corner. Unlike the first lock-in region, the vortex does not interact with the trailing edge after shedding off. As a result, the fluctuating pressure on the trailing edge is relatively small. The rectangular cylinder moves upward to position C, the upper surface vortex U1 locates at the position corresponding to the maximum positive contribution value and does positive work; the lower surface vortex D2 locates at the position corresponding to the maximum negative contribution value and does negative work. Then the rectangular cylinder moves upward to position D, and the upper surface vortex sheds from the trailing edge corner and a new vortex U2 generates at the leading edge. Finally, the rectangular cylinder moves downward back to position A, and the upper surface vortex U2 and the lower surface vortex D2 move to the original positions of U1 and D1, respectively. Then a new cycle begins.

According to the simplified vortex model theory (Hu et al., 2022), the actual reduced velocity U_A^* of the vertical VIVs is expressed as:

$$U_A^* = \frac{U}{f_h B_A} = \frac{U}{n U_{\text{vortex}}} = \frac{1}{n R_v} \quad (6)$$

where B_A is the vortex drifting distance; U_{vortex} is the drift velocity of the separated vortex; R_v denotes the ratio of the drift velocity of the separated vortex to the oncoming wind velocity, and is about 0.6 (Hu et al., 2021). n is the number of periods required for the separated vortex to drift from the leading edge to the trailing edge. For the first and second lock-in regions, approximately take $R_v=0.6$ and take $n = 2$ and 1 , respectively. Then the calculated U_A^* for the two lock-in regions are 0.83 and 1.67. For the 6:1 rectangular cylinder, take $B_A=B=6D$, and then the calculated reduced velocity V_r ($V_r=U/f_h D$) are 5.0 and 10.0, respectively, which are very close to the onset wind velocities of the two lock-in regions in Fig. 4.

In the two lock-in regions, the actual flow field from the CFD simulation (Fig. 10(b) and Fig. 11(b)) corresponds well with the simplified vortex model, which proves that it is feasible to deduce the key flow field characteristics based on the wind pressure distribution characteristics on the structure surface (especially the simple harmonic curve distribution of the contribution value and the linear increasing trend of the phase lag), and reveal the VIV mechanism from the joint viewpoint combining surface wind pressure distribution and flow pattern characteristics.

4.2. Flow pattern under dynamic mode decomposition

In the above discussion, the flow field characteristics around the rectangular cylinder corresponding to the two lock-in regions have been preliminarily elucidated. However, more information could be hidden in the complex vortex structure of the original flow field. DMD method, proposed by Schmid (2010), is a useful way to get the single-frequency flow mode and corresponding temporal and spatial characteristics of the flow field. This method is suitable for further exploring the evolution process of vortex structure in the periodic flow, and describe the underlying physical mechanisms captured in the data sequence.

Schmid (2010) and Kutz et al. (2016) introduced the mechanism and validation of DMD method in detail. In the present study, this method is used in the following steps: Original simulation results of velocity field are sampled with a constant time interval δ_t , and the snapshot information of t moments are recorded to establish the snapshot matrixes $X_1 = \{\nu_1, \nu_2, \nu_3, \dots, \nu_{t-1}\}$ and $X_2 = \{\nu_2, \nu_3, \nu_4, \dots, \nu_t\}$. It is assumed that there is a linear relationship between the flow field at the i^{th} time and the flow field at the next time, that is: $X_2 = A X_1$. Take the singular value decomposition (SVD) of X_1 , and for calculation convenience, only the first r modes with higher energy are retained to approximate the original flow field. It can be solved that:

$$A = X_2 V S^{-1} U^* \quad (7)$$

$$\tilde{A} = U^* A U = U^* (X_2 V S^{-1} U^*) U = U^* X_2 V S^{-1} \quad (8)$$

where $\tilde{A}$ is the similar matrix of A , U is the orthogonal basis of the velocity field, S represents the energetic importance of each mode, V returns time series corresponding to each mode, and $*$ represents conjugate transposition.

The order of $\tilde{A}$ is only $r \times r$, so that it is efficient to get its eigenvalues and eigenvectors:

$$[W, \Lambda] = \text{eig}(\tilde{A}) \quad (9)$$

A owns the same eigenvalues $\lambda_i = \text{diag}(\Lambda)$ with its similar matrix $\tilde{A}$. The eigenvalues and the eigenvectors of the DMD modes can then be calculated (Tu et al., 2013) by:

$$\omega_i = \frac{\log(\lambda_i)}{\delta t} \quad (10)$$

$$\hat{\Phi} = X_2 V \Lambda^{-1} W \quad (11)$$

The real part of ω_i denotes the growth rate of the i^{th} mode, while the imaginary part denotes the oscillating frequency.

The time-varying flow field modes can be approximately reconstructed with the former r -order modes:

$$\nu_t = \sum_{k=1}^r \phi_k \exp(\omega_k t) b_k = \Phi \exp(\Lambda) b \quad (12)$$

where, ν_t is the snapshot of the velocity field at the t moment, b represents the projection of the mode onto the snapshot, i.e. the amplitude.

DMD method is used to explore the vortices evolution dominating the global instability. The total number of snapshots under each velocity is 1255, and the time interval between two consecutive snapshots is 0.004s. That is, the velocity field data from about 33 vibration periods is used to conduct DMD. The modes of the velocity field are extracted and sorted by their amplitude, which indicating their contribution to the original flow field (Jovanović et al., 2014). When the sum of energy account of the first several modes exceeds 99 % of the total of the flow field, the remaining modes will be discarded because their energy accounts are very small.

For the two peak amplitude points, the eigenvalues λ_i of the first several modes are shown in Fig. 12(a) and (b), respectively. The eigenvalues of most of the modes fall on the unit circle, indicating that the modes are periodically stable with time. Eigenvalues in the interior of the unit circle are strongly damped and only influence early stages of time evolution (Jovanović et al., 2014). The distribution of the mode eigenvalue is symmetry about the real axis, and there is a conjugate mode for each frequency non-0 mode. Each pair of conjugate modes has the same flow pattern, and only the modes with positive frequencies are studied. The first 4 non-negative modes of the flow fields at the two peak amplitude points as well as their growth-ratio and contributions are shown in Table 2, respectively. The frequencies of the first 4 non-negative modes of the two lock-in regions are similar. The frequency of the first mode is 0, and this mode corresponds to the time-averaged flow field, which does not change over time. The frequencies of the following three modes are about once, twice and three times of the natural frequency of the rectangular cylinder (f_h), respectively. The contribution of the second mode (whose frequency is about equal to the natural frequency) is much larger than the following two modes. For comparison, the amplitude spectrums of the general VEF at the two peak amplitude points are shown in Fig. 13. The general VEF of the second peak amplitude point has obvious components corresponding to twice and three times of natural frequency. This phenomenon also exists at the first lock-in region, however, it is not very significant.

The flow field corresponding to the first modes at the two peak amplitude points are shown in Fig. 14. The streamline is marked with white lines and arrows in these figures. According to Fig. 14, the flow fields of Mode 1 in the two lock-in regions are quite similar, corresponding to their similar mean pressure coefficient distribution characteristics (Fig. 5). In the time-averaged flow field, there are two large vortices in the front area of the upper and lower surfaces of the rectangular cylinder, and these two vortices rotate in opposite directions. Compared with Fig. 5, it can be seen that at the location where this pair of vortices exist, the mean pressure coefficients on the upper and lower surface show relatively large negative values, indicating that the surface is under suction; and on the downstream area of the vortices, the suction values of the mean pressure significantly decrease. Besides, on the trailing edge of the rectangular cylinder, there are also a pair of smaller vortices with opposite rotation directions.

The flow field corresponding to the following three DMD modes (Mode 2, Mode 3 and Mode 4) of the two peak amplitude points are shown in Fig. 15 and Fig. 16, respectively. For the first lock-in region, in Mode 2, it can be clearly seen that there are four vortices on the upper and lower surfaces of the rectangular cylinder, respectively, and there are also obvious vortices in the wake. Among the four vortices on each surface, two vortices rotate clockwise and two vortices rotate counterclockwise, with adjacent vortices rotating in opposite directions. At the same position, the upper surface vortex and the lower surface vortex rotate in the same direction. For observation convenience, the vortices on the rectangular cylinder surfaces in Fig. 15(a) are abstracted and simplified into the schematic diagram in Fig. 15(b). Compared with Fig. 10, it can be seen that each pair of adjacent vortices on each surface in Mode 2 corresponds to one vortex in the simplified vortex model. Zhang et al. (2023) also found similar phenomenon that the leading-edge vortex shedding in the original flow is equivalent to the alternative shedding of the counter-rotating vortex pairs in the DMD mode. Among the modes whose frequency is not equal to 0, the contribution of Mode 2 is the largest, which means that the flow field of this mode plays an important role in the occurrence of VIV. In Mode 3, the streamlines near the surfaces of the rectangular cylinder are not very uniform, there are multiple vortices drifting on each surface, but it's difficult to identify the number of vortices. Pairs of vortices can also be clearly observed in the wake. The upper surface vortices always rotate in the opposite direction with the lower surface vortices in Mode 3. In Mode 4, significant vortices can also be observed in the wake. In Mode 3 and Mode 4, vortices in the wake appear to be more significant than vortices on the surfaces.

The flow field of the second peak amplitude point (Fig. 16) is quite differ from that of the first peak amplitude point (Fig. 15). There are only two vortices on the upper and lower surfaces of the rectangular cylinder. The number of vortices on each surface at the second peak amplitude point is half of that of the first peak amplitude point, and this corresponds well with the above discussion in Section 4.1. Similar to Fig. 15(a), in Mode 2, the vortices on the same surface rotate in opposite directions, and the upper surface vortex and the lower surface vortex at the same position rotate in the same direction. In Mode 3 and Mode 4, significant vortices can also be observed

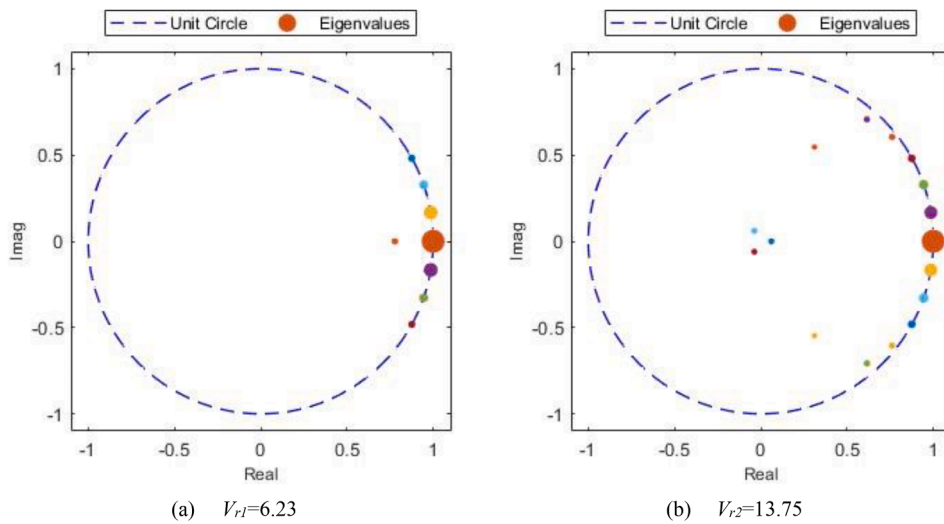


Fig. 12. Dynamic mode decomposition eigenvalues of velocity field.

Table 2

DMD modes at peak amplitude points of two lock-in regions.

The first peak amplitude point				The second peak amplitude point			
Mode	Frequency	Growth-ratio	Contribution	Mode	Frequency	Growth-ratio	Contribution
1	0 Hz	$-1.18\text{e-}6$	77.71 %	1	0 Hz	$-1.01\text{e-}7$	79.58 %
2	6.7 Hz	$7.77\text{e-}7$	9.12 %	2	6.7 Hz	$-1.02\text{e-}4$	6.80 %
3	13.3 Hz	$-1.57\text{e-}3$	1.36 %	3	13.3 Hz	$-8.10\text{e-}3$	1.56 %
4	20.0 Hz	$-2.22\text{e-}2$	0.49 %	4	20.0 Hz	$-6.15\text{e-}2$	0.78 %

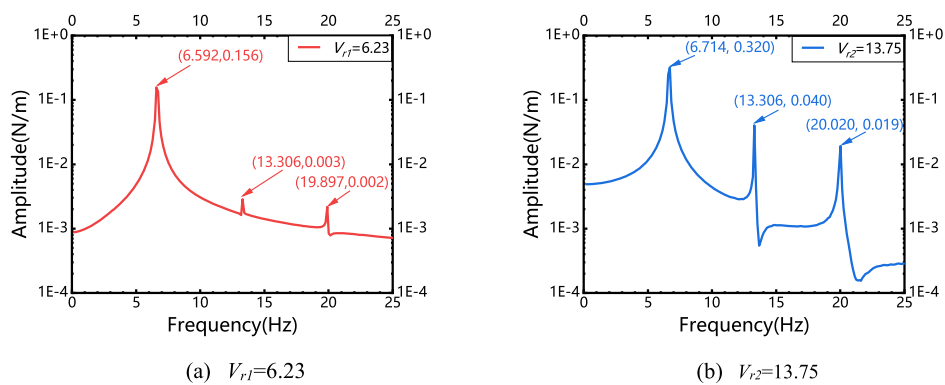


Fig. 13. Amplitude spectra of VEF at peak amplitude points of two lock-in regions.

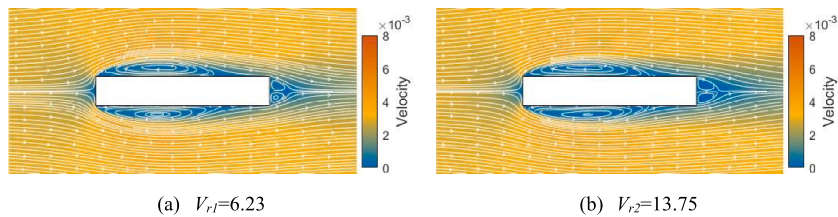


Fig. 14. The first DMD mode at peak amplitude point of two lock-in regions.

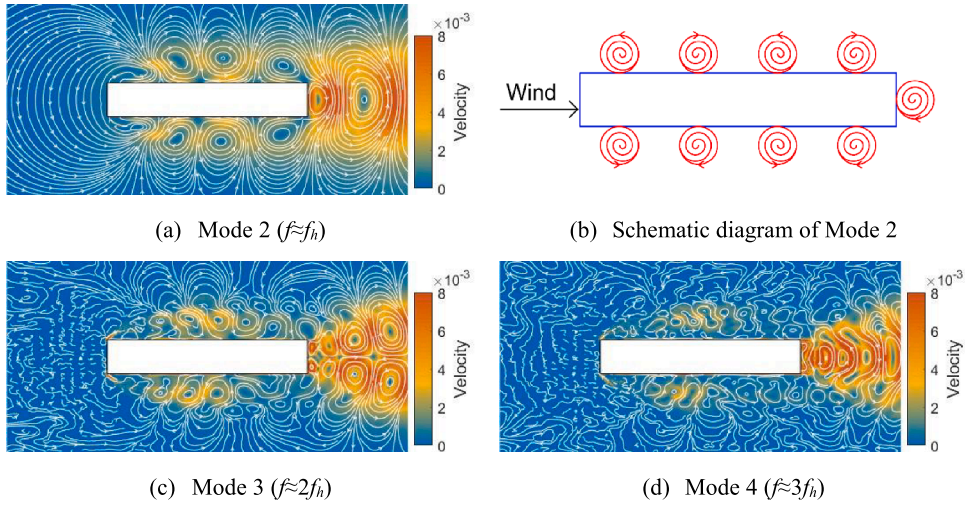


Fig. 15. Velocity fields and surface vortices of main DMD modes of the first lock-in region.

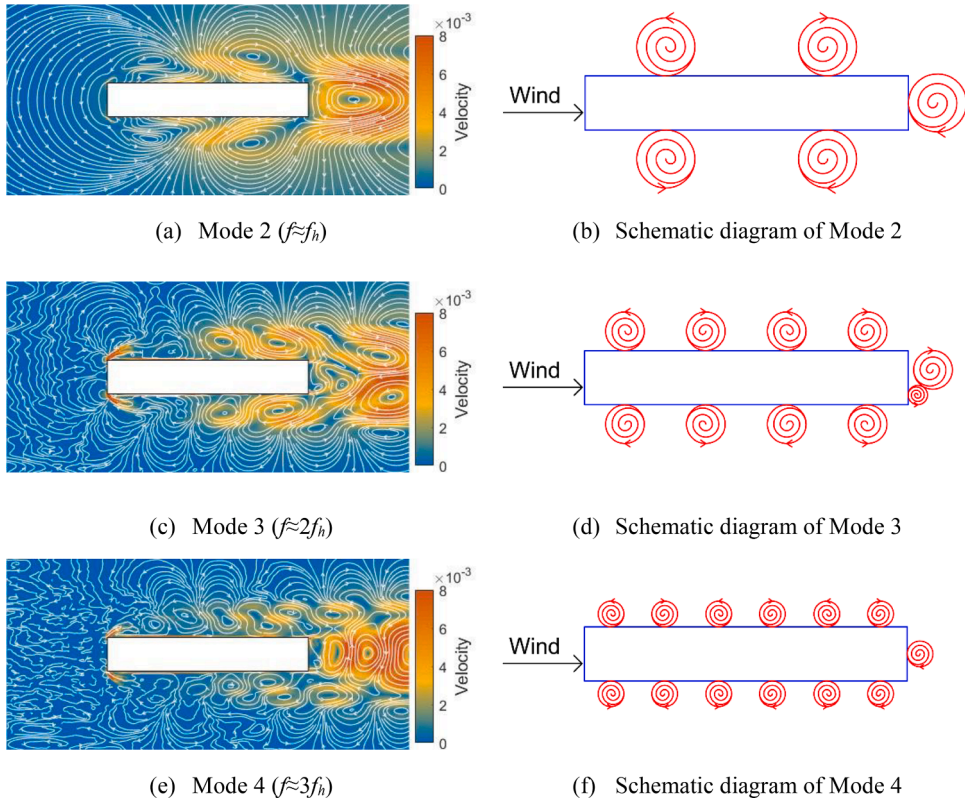


Fig. 16. Velocity field of main DMD modes at the second peak amplitude point.

on the upper and lower surfaces. In Mode 3, there are four vortices drifting on each surface, twice of that in Mode 2, and at the same time, the frequency of Mode 3 is also twice of the frequency of Mode 2. In Mode 4, there are 6 vortices drifting on the rectangular cylinder surfaces, three times of that in Mode 2. In other words, with the increasing of the mode frequency, multiple vortices appear on the rectangular surfaces, which may be the source of the nonlinearity of the general VEF. The upper surface vortices and the lower surface vortices at the same position rotate in the opposite direction in Mode 3. This makes the flow pattern in Mode 3 of the second lock-in region different with the flow pattern in Mode 2 of the first lock-in region, although the numbers of vortices on the surfaces in these two modes are the same. The vortices in the DMD modes of the second lock-in region are also abstracted and simplified into the schematic diagram in Fig. 16(b), (d), (f). Compared with Fig. 11, it can be seen that each vortex in the simplified vortex model also

corresponds to a pair of vortices with opposite rotation directions in Mode 2.

To sum up, by DMD method, the main modes of the flow field are extracted. For both the peak amplitude points of the two lock-in regions, the frequencies of the first three non-zero frequency DMD modes (Mode 2, Mode 3, Mode 4) correspond to once, twice, and three times of the natural frequency of the rectangular cylinder, respectively, and the energy contribution of the modes corresponding to the natural frequency is much higher than the following modes. Significant vortex drifting phenomenon on the cylinder surface is observed in these three modes, especially in the second lock-in region, indicating the important role of this phenomenon in the occurrence of VIVs. The vortices at the trailing edge are also obvious, especially in Mode 3 and Mode 4 of the first lock-in region.

5. Conclusions

A numerical simulation of a 6:1 rectangular cylinder was conducted to elucidate the multiple-order vertical VIV mechanism. Within the investigated wind velocity range, the 6:1 rectangular cylinder experienced two lock-in regions. The wind pressure distribution characteristics and flow patterns were investigated and compared to reveal the differences between the two lock-in regions. Then DMD method was used to further clarify the flow pattern characteristics around the rectangular cylinder in different modes. The main conclusions are as follows:

- 1) The mean pressure coefficient on the surfaces of the rectangular cylinder shows similar distribution characteristics in the two lock-in regions, while the fluctuating pressure coefficient, correlation coefficient and contribution value show significant differences: two peak values are observed on each surface in the first lock-in region, but only one in the second lock-in region. This indicates that there are two and one vortices on each surface in the first and second lock-in region, respectively.
- 2) There are significant vortices periodically drifting on the surfaces of the rectangular cylinder in both the two lock-in regions, however, it takes about two and one period for the vortices generated from the leading edge to drift to the trailing edge in the first and second lock-in regions, respectively. A simplified vortex model was built basing on the distribution trend of contribution value and phase lag to describe this phenomenon of periodic vortex drifting, which corresponds well with the actual flow field.
- 3) The first three non-zero DMD mode frequencies are close to once, twice, and three times of the natural frequency of rectangular cylinder, respectively. The number of vortices on the upper and lower surfaces in the first lock-in region is twice of those in the second lock-in regions in the first non-zero frequency DMD mode. Multiple vortices are observed to drift on the surfaces in the high-order DMD modes with multiple frequencies, especially in the second lock-in region, indicating the importance of vortex drifting phenomenon in the occurrence of VIVs.

This investigation provides a comprehensive method which reveals the VIV mechanism from both the aspects of the surface pressure characteristics and the surrounding flow patterns. The key flow field patterns closely related to VIV can be speculated based on the surface pressure distribution characteristics, then the simplified vortex model can be established to reveal the VIV mechanism. The correctness of the speculated flow pattern and the simplified vortex model can be verified by introducing actual flow field and the flow field of DMD modes. This method provides a reference for the investigation of VIV mechanism of structures with more complex configurations.

CRedit authorship contribution statement

Lin Zhao: Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Tingshu Han:** Writing – original draft, Methodology, Formal analysis, Data curation. **Chuanxin Hu:** Writing – review & editing, Methodology. **Song Luo:** Methodology, Formal analysis, Data curation. **Qiang Zhou:** Writing – review & editing, Supervision. **Wei Cui:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Bruno, L., Fransos, D., Coste, N., Bosco, A., 2010. 3D flow around a rectangular cylinder: a computational study. *J. Wind Eng. Ind. Aerod.* 98 (6–7), 263–276.
- Bruno, L., Salvetti, M., Ricciardelli, F., 2014. Benchmark on the aerodynamics of a rectangular 5:1 cylinder: an overview after the first four years of activity. *J. Wind Eng. Ind. Aerod.* 126, 87–106.
- Gao, G.Z., Zhu, L.D., 2015. Nonlinearity of mechanical damping and stiffness of a spring-suspended sectional model system for wind tunnel tests. *J. Sound Vib* 355, 369–391.
- Ge, Y.J., Zhao, L., Cao, J.X., 2022. Case study of vortex-induced vibration and mitigation mechanism for a long-span suspension bridge. *J. Wind Eng. Ind. Aerod.* 220, 1–13.
- Guéniat, F., Pastur, L., Lusseyran, F., 2014. Investigating mode competition and three-dimensional features from two-dimensional velocity fields in an open cavity flow by modal decompositions. *Phys. Fluids* 26, 085101.
- Hu, C.X., Zhao, L., Ge, Y.J., 2018. Time-frequency evolutionary characteristics of aerodynamic forces around a streamlined closed-box girder during vortex-induced vibration. *J. Wind Eng. Ind. Aerod.* 182, 330–343.
- Hu, C.X., Zhao, L., Ge, Y.J., 2021. A simplified vortex model for the mechanism of vortex-induced vibrations in a streamlined closed-box girder. *Wind Struct* 32, 309–319.
- Hu, C.X., Zhao, L., Ge, Y.J., 2022. Multiple-order vertical vortex-induced vibration mechanism of a typical streamlined closed-box girder. *J. Wind Eng. Ind. Aerod.* 227, 105066.
- Hu, C.X., Zhao, L., Wang, X.L., Ge, Y.J., 2024. Spatio-temporal distribution of aerodynamic forces and their associated vortex drift patterns around a closed-box girder during torsional vortex-induced vibration. *Engineering Structures* 302, 117459.
- Jovanović, M.R., Schmid, P.J., Nichols, J.W., 2014. Sparsity-promoting dynamic mode decomposition. *Phys. Fluids* 26 (2), 024103.
- Komastus, S., Kobayashi, H., 1980. Vortex-induced oscillation of bluff cylinders. *J. Wind Eng. Ind. Aerod.* 6, 335–362.
- Kubo, Y., Hirata, K., Mikawa, K., 1992. Mechanism of aerodynamic vibrations of shallow bridge girder sections. *J. Wind Eng. Ind. Aerod.* 42, 1297–1308.
- Kutz, J.N., Brunton, S.L., Brunton, B.W., Proctor, J.L., 2016. Dynamic mode decomposition: data-driven modeling of complex systems. *Soc. Industr. Appl. Mathem.*
- Laima, S.J., Li, H., Chen, W.L., Ou, J.P., 2018. Effects of attachments on aerodynamic characteristics and vortex-induced vibration of twin-box girder. *J. Fluid Struct.* 77, 115–133.
- Li, W.L., Patruno, L., Niu, H.W., Miranda, S.D., Hua, X.G., 2021. Aerodynamic admittance of a 6:1 rectangular cylinder: a computational study on the role of turbulence intensity and integral length scale. *J. Wind Eng. Ind. Aerod.* 218, 104738.
- Li, X.T., Lyu, Z., Kou, J.Q., Zhang, W.W., 2019. Mode competition in galloping of a square cylinder at low Reynolds number. *J. Fluid Mech.* 867, 516–555.
- Liu, S.Y., Lin, Z., Fang, G.S., Hu, C.X., Ge, Y.J., 2021. Investigation on aerodynamic force nonlinear evolution for a central-slotted box girder under torsional vortex-induced vibration. *J. Fluid Struct.* 106, 103380.
- Luo, X., Kareem, A., 2021. Dynamic mode decomposition of random pressure fields over bluff bodies. *J. Fluid Mech.* 147 (4), 1–20.
- Lusseyran, F., Guéniat, F., Basley, J., Douay, C.L., Pastur, L.R., Faure, T.M., Schmid, P.J., 2011. Flow coherent structures and frequency signature: application of the dynamic modes decomposition to open cavity flow. *J. Phys.* 318 (4), 042036.
- Mills, R., Sheridan, J., Hourigan, K., 2003. Particle image velocimetry and visualization of natural and forced flow around rectangular cylinders. *J. Fluid Mech.* 478, 299–323.
- Matsumoto, M., Shirato, H., Aaraki, K., Haramura, T., Hashimoto, T., 2003. Spanwise coherence characteristic of surface pressure field on 2D bluff bodies. *J. Wind Eng. Ind. Aerod.* 91, 155–163.
- Nakamura, Y., Nakashima, M., 1986. Vortex excitation of prisms with elongated rectangular, H and I cross-sections. *J. Fluid Mech.* 163, 149–169.
- Nakamura, Y., Ohya, Y., Tsuruta, H., 1991. Experiments on vortex shedding from flat plates with square leading and trailing edges. *J. Fluid Mech.* 222, 437–447.
- Nakamura, Y., Ohya, Y., Ozono, S., Nakayama, R., 1996. Experimental and numerical analysis of vortex shedding from elongated rectangular cylinders at low Reynolds numbers $200\text{--}10^3$. *J. Wind Eng. Ind. Aerod.* 65, 301–308.
- Nguyen, D.T., Hargreaves, D.M., Owen, J.S., 2018. Vortex-induced vibration of a 5:1 rectangular cylinder: a comparison of wind tunnel sectional model tests and computational simulations. *J. Wind Eng. Ind. Aerod.* 175, 1–16.
- Ohya, Y., Nakamura, Y., Ozono, S., Tsuruta, H., Nakayama, R., 1992. A numerical study of vortex shedding from flat plates with square leading and trailing edges. *J. Fluid Mech.* 236, 445–460.
- Shang D.Y., 2015. Experimental study on model scale effect of vortex-induced vibration for 1:6 rectangular cross-sections [dissertation]. *China: Hunan University*. [In Chinese].
- Schmid, P.J., 2010. Dynamic mode decomposition of numerical and experimental data. *J. Fluid Mech.* 656, 5–28.
- Shang, J.M., Zhou, Q., Alam, M.M., Liao, H.L., Cao, S.Y., 2019. Numerical studies of the flow structure and aerodynamic forces on two tandem square cylinders with different chamfered-corner ratios. *Phys. Fluids* 31, 075102.
- Shiraishi, N., Matsumoto, M., 1983. On classification of vortex-induced oscillation and its application for bridge structures. *J. Wind Eng. Ind. Aerod.* 14 (1), 419–430.
- Tu, J.H., Rowley, C.W., Luchtenburg, D.M., Brunton, S.L., Kutz, J.N., 2013. On dynamic mode decomposition: theory and applications. *J. Comput. Dynam.* 1 (2), 391–421.
- Wu, B., Zhang, L.L., Yang, Y., Liu, L.J., Li, H.H., 2017. Investigation and control of VIVs with multi-lock-in regions on wide flat box girders. *J. Control Sci. Eng.* 2017, 1–17.
- Zhang, Q., Liu, Y., Wang, S., 2014. The identification of coherent structures using proper orthogonal decomposition and dynamic mode decomposition. *J. Fluids Struct.* 49, 53–72.
- Zhang, Z.B., Xu, F.Y., 2020. Spanwise length and mesh resolution effects on simulated flow around a 5:1 rectangular cylinder. *J. Wind Eng. Ind. Aerod.* 202, 104186.
- Zhang, Z.B., Kareem, A., Xu, F.Y., Jiang, H.Y., 2023. Global instability and mode selection in flow fields around rectangular prisms. *J. Fluid Mech.* 955, A19.
- Zhao, L., Cui, W., Shen, X.M., Xu, S.Y., Ding, Y.J., Ge, Y.J., 2022. A fast on-site measure-analyze-suppress response to control vortex-induced-vibration of a long-span bridge. *Structures* 35, 192–201.
- Zhou, Q., Shang, J.M., Alam, M.M., Li, H., 2024. Effects of chamfering and spacing on aerodynamics of two tandem cylinders. *Phys. Fluids* 36, 115176.